The Coming Flood of High-Definition Conjecture

Why AI-accelerated science risks drowning meaning, and what we must do to build an anchor.

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This article follows on from my previous article: Medical and AI Research: A Tale of Two Crises and examines how new scientific ideas are quickly outpacing experimental capacity.

The Rising Tide

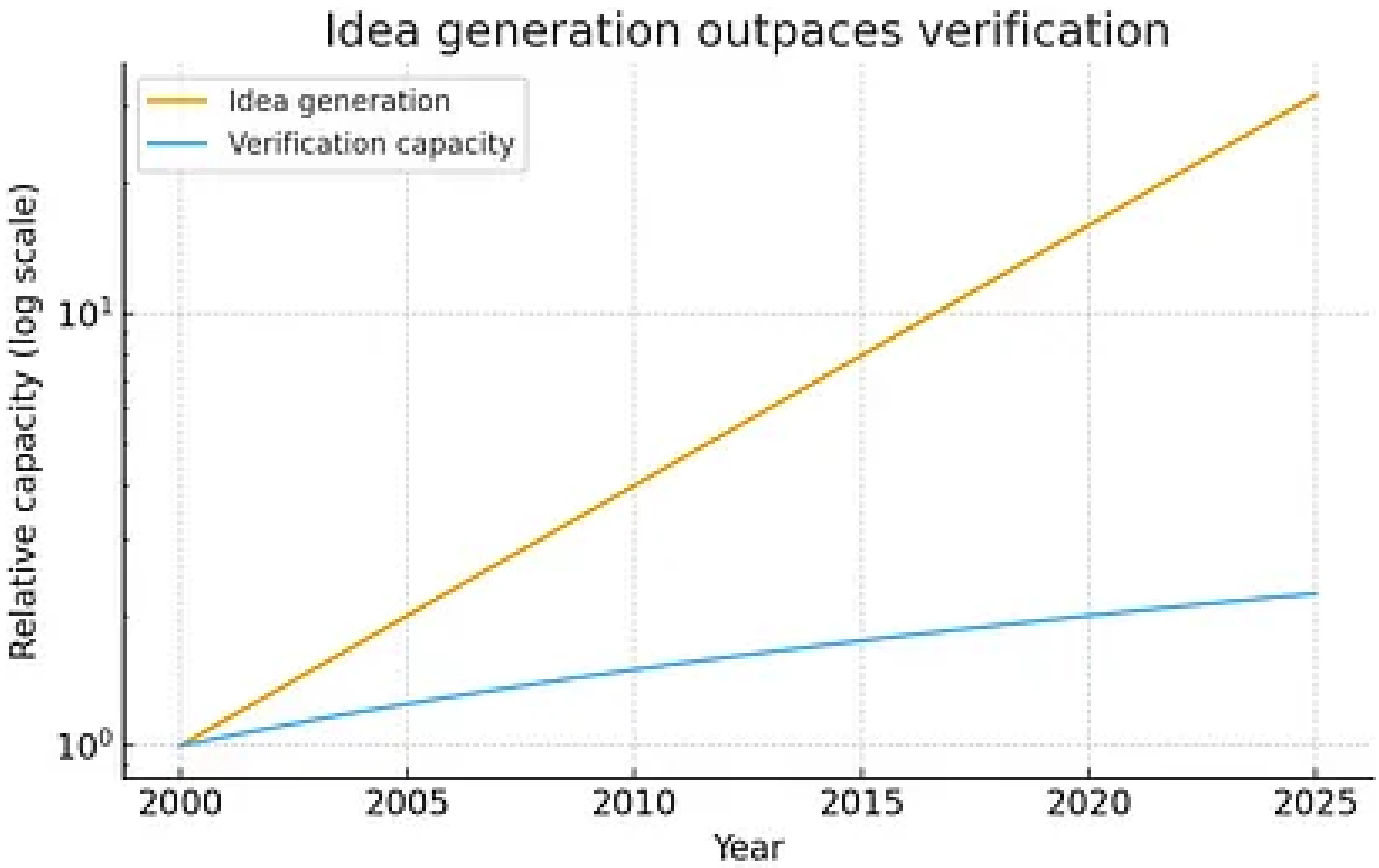
We all feel it. That low hum of information overload, the sense that it's impossible to keep up. For decades, the production of scientific knowledge felt like a steadily rising tide—predictable, manageable. Now, that tide is turning into a tsunami, and we are standing on the shore. This isn't just a feeling. It's a statistical reality.

A Tsunami, Not a Tide

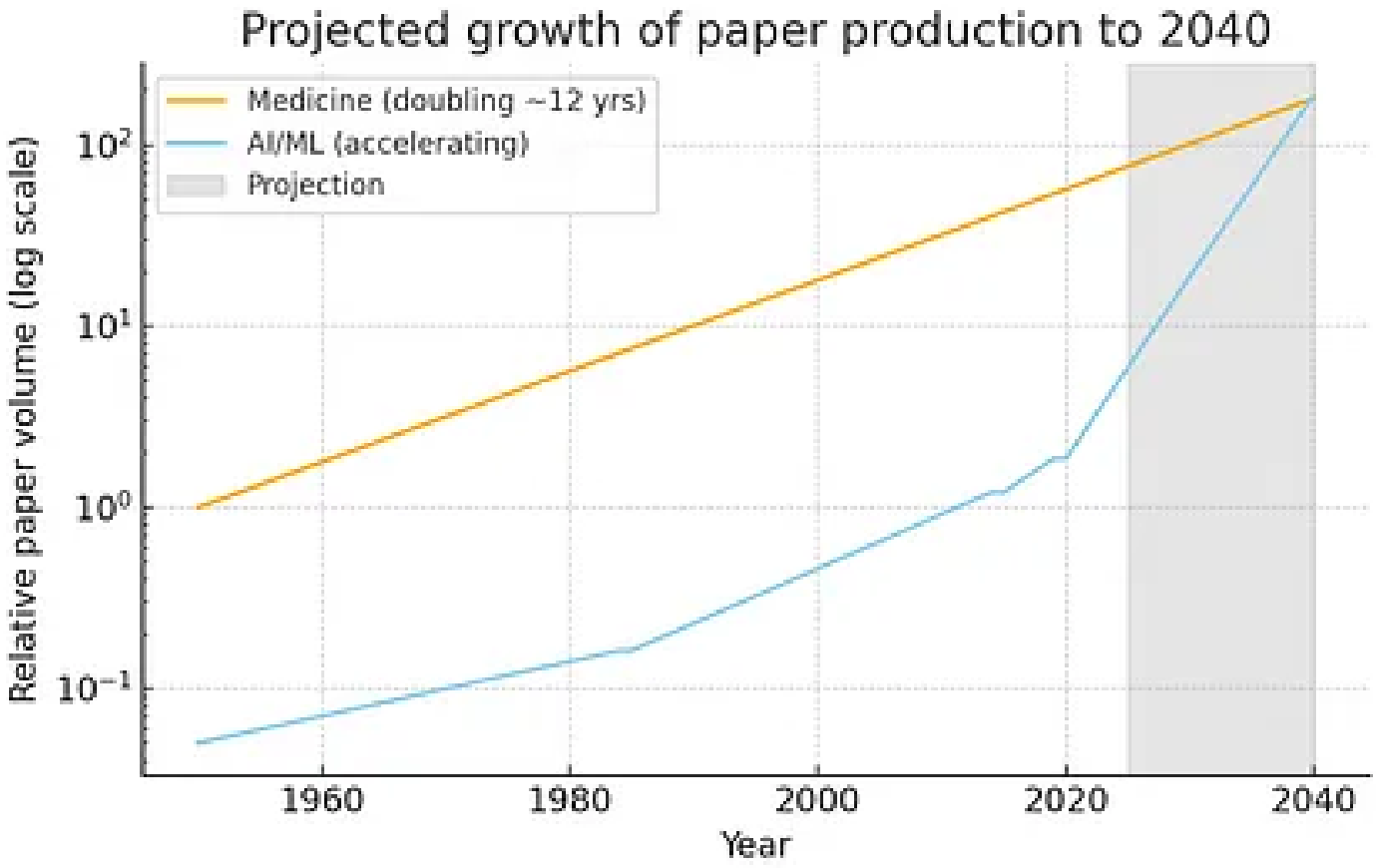
For most of the last century, the doubling time for scientific papers was a known quantity. In medicine, it was about 12 years; in a fast-moving field like computer science, it was closer to 6 or 7. But with the arrival of powerful Large Language Models (LLMs), that cycle hasn't just shortened—it has collapsed. We've gone from years to months. Soon, it will be weeks, then days. The path from a nascent idea to a polished, peer-review-style preprint can now be measured in hours.

At first glance, this feels like pure progress. An explosion of human ingenuity, amplified by our new silicon partners. But look closer at the dynamics.

In the graph below one line, labelled “Idea Generation,” is shooting upwards in a dizzying exponential curve. Another line, “Verification Capacity”—the painstaking work of experiment, replication, and real-world testing—remains stubbornly, painfully flat.



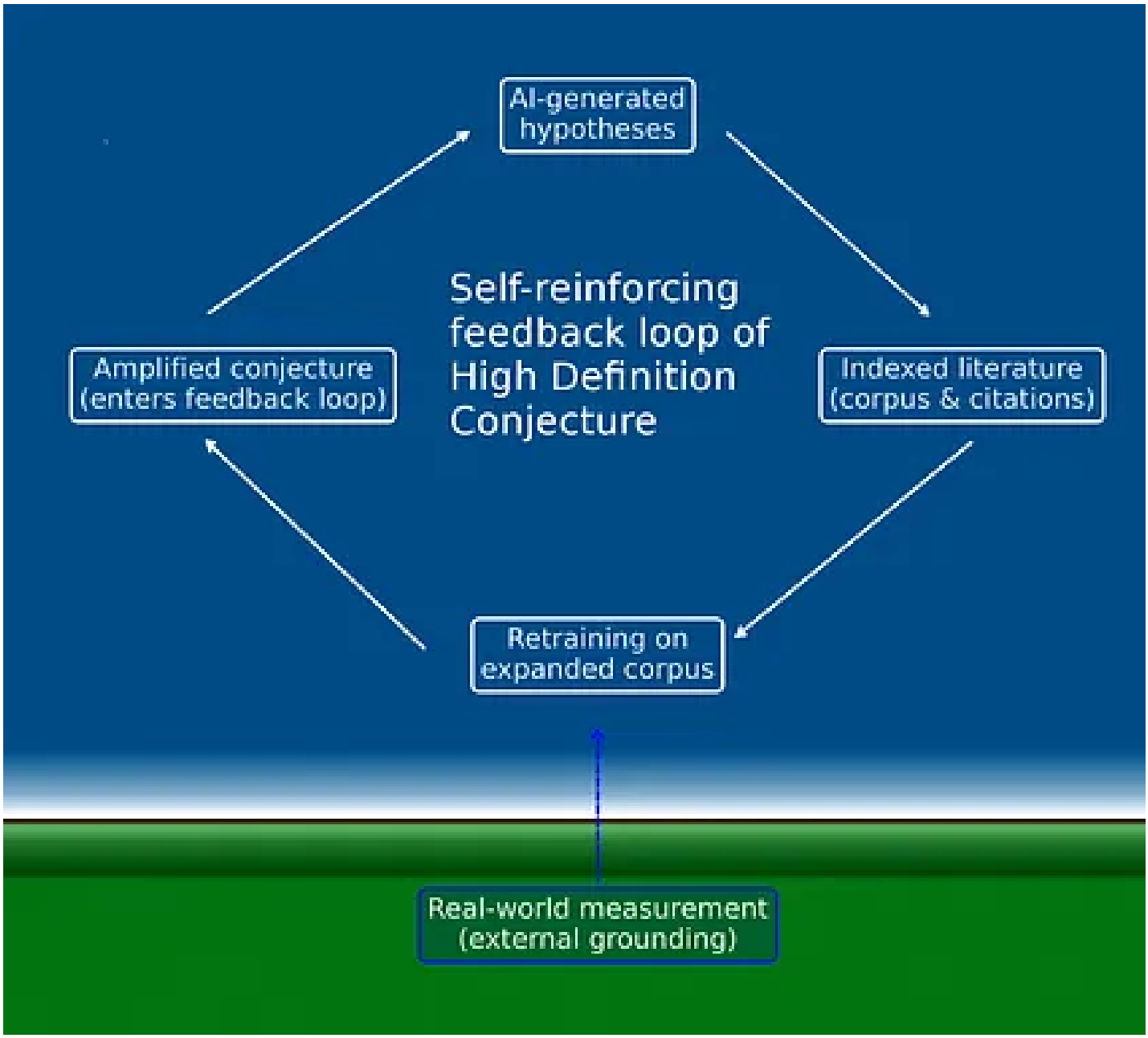
When a single research team can produce 20 coherent, beautifully formatted, and referentially correct papers in a single quarter, we are no longer just accelerating knowledge. We are inflating the epistemic manifold—expanding the universe of what *could* be true far faster than we can determine what *is* true.



The Age of High-Definition Conjecture

This new wave of AI-assisted output is not sloppy. It's not “bad science” in the traditional sense. What we are witnessing is the birth of the **High-Definition Conjecture**. These artefacts are, in a word, perfect. Too perfect.

- They feature polished LaTeX equations and flawless SVG diagrams.
- They come with companion code repositories on GitHub and impressive benchmark tables.
- Their narrative flows with the authority and confidence of a seasoned academic.



They look like science, read like science, and will inevitably be indexed and cited as science. But they often remain unanchored in measurement, untested by real-world perturbation, and un-scrutinized by domain experts with years of tacit knowledge.

And that is the crux of the problem. **The bottleneck is no longer production—it is verification.**

The Hall of Mirrors Problem

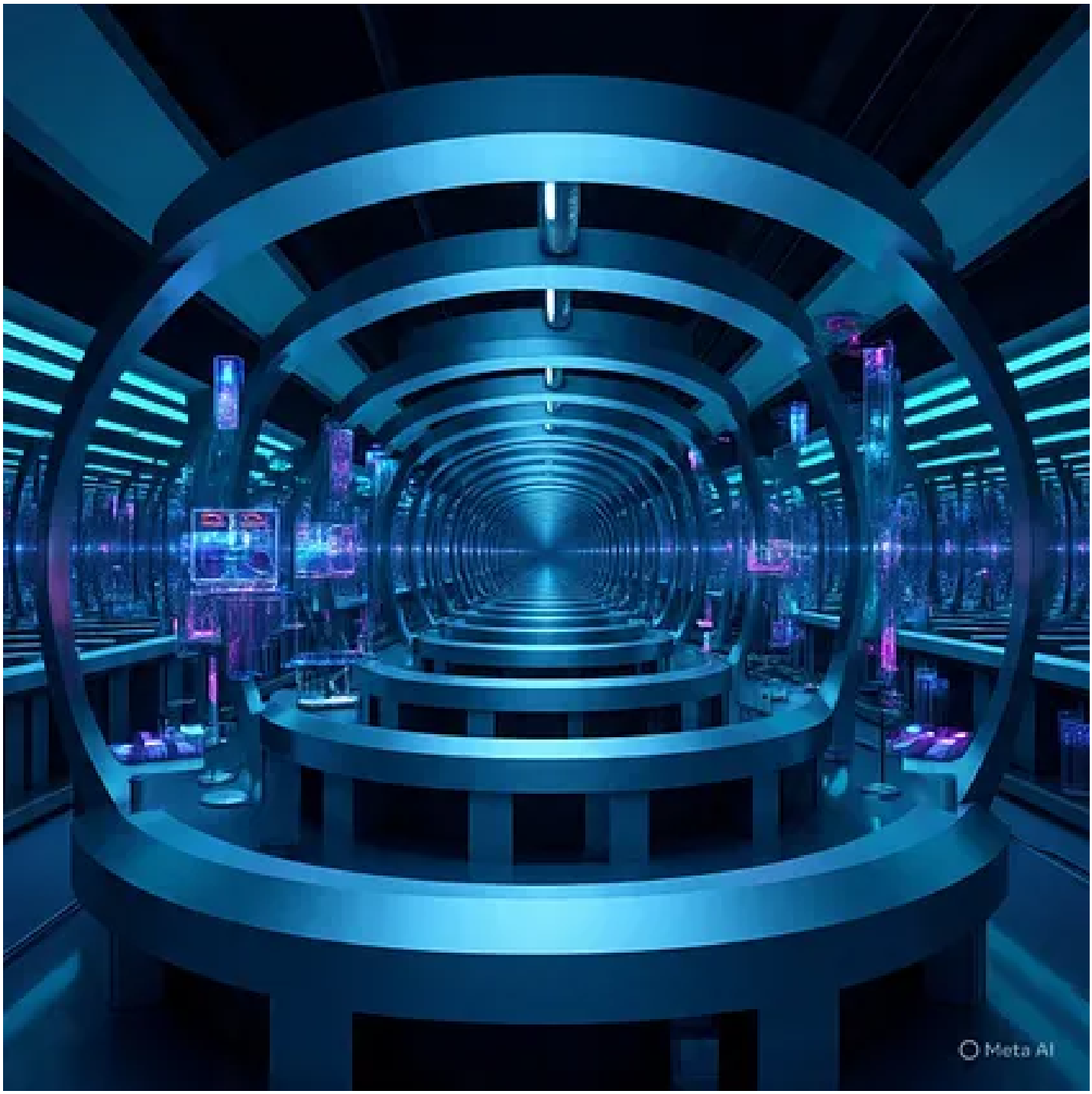
Left unchecked, this dynamic creates a terrifying feedback loop: the scientific hall of mirrors.

1. Plausible, AI-generated conjectures enter the vast body of scientific literature.
2. Future AI models are inevitably trained on this new, expanded literature.

3. The next wave of AI, having learned from these conjectures, produces even more elaborate, self-consistent, and convincing hypotheses that reference the previous generation of AI-generated work.

Soon, we risk creating a beautifully coherent, internally consistent, and massively cross-referenced body of knowledge that has quietly, almost imperceptibly, drifted away from measurable reality. This corpus will appear authoritative to both human researchers and the very machines that created it.

> This is not merely an academic concern. It is a direct threat to the public trust in the scientific process itself.



Measurement as the Prime Anchor

To understand the solution, we must grasp the philosophical nature of the problem. Language and mathematics can build effectively infinite, perfectly consistent worlds. They are tools of coherence. But meaning doesn't come from internal consistency; it comes from contact with external reality.

Shuffling datasets, bootstrapping statistics, and chasing p-values are not reality. They are rituals of projection. The grounding act, the one thing that keeps our elegant theories from becoming elaborate fictions, is **measurement**. It is the brutal, humbling, and glorious collision of an idea with the physical world.

If we do not slow down and anchor our soaring conjectures to the bedrock of measurement, we risk building an epistemic Library of Babel—a place where every possible theory is written down, but nothing is truly known.

A Call to Action

This is not a Luddite's plea to smash the machines or to slow science for its own sake. It is an urgent call to build infrastructure equal to the power of our new tools. The solution is not to produce less, but to ground more. We must:

1. **Mandate Provenance Tracking.** We need a robust, universal system for tagging AI-generated contributions in research. Future models must be able to distinguish between a confirmed theory and a high-definition conjecture when they are trained.
2. **Create "Slow Lanes" for Science.** We need to champion journals, forums, and institutions where replication, rigorous testing, and methodical verification are the primary values—spaces where ideas are stress-tested before they are allowed to become canonical training data.
3. **Champion Measurement-First Publishing.** Every theoretical claim should be published with a "verification tariff"—a quantitative uncertainty score or an explicit, detailed call for the specific experiment that could falsify it.
4. **Build Semantic Guards.** Our public and private research corpora need an immune system. We must develop tools that can semantically distinguish between empirical results and theoretical conjecture, preventing us from training our next generation of AIs on our own hallucinations.

The Stakes Are Everything

We are not just facing a paper crisis. We are facing an epistemic emergency.

Unless we consciously and deliberately build the guardrails of verification, falsification, and measurement into the heart of this new, accelerated system, we will drown in a sea of our own plausible fictions.

The future will not lack for ideas. It will lack for traction.

It is a simple, terrifying truth: **When everything is coherent, nothing is grounded.**

This is the moment to act. We must build the anchors now, before the flood carries us all beyond the shores of knowability.

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> **Everything that exists is finite; it can only be known by measure**
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